American Computer Science League

2025-26 Finals ● Short Problems Solutions ● Senior Division

1. Boolean Algebra

$\overline{\overline{(A + \overline{AB(B + C)})}}(\overline{CD} + D) = (A + \overline{AB(B + C)}) + \overline{(\overline{CD} + D)} =$
$A + \overline{A} + \overline{B} + \overline{B}\,\overline{C} + CD\,\overline{D} = 1 + 0 = 1$ This is TRUE for all 16 of them.

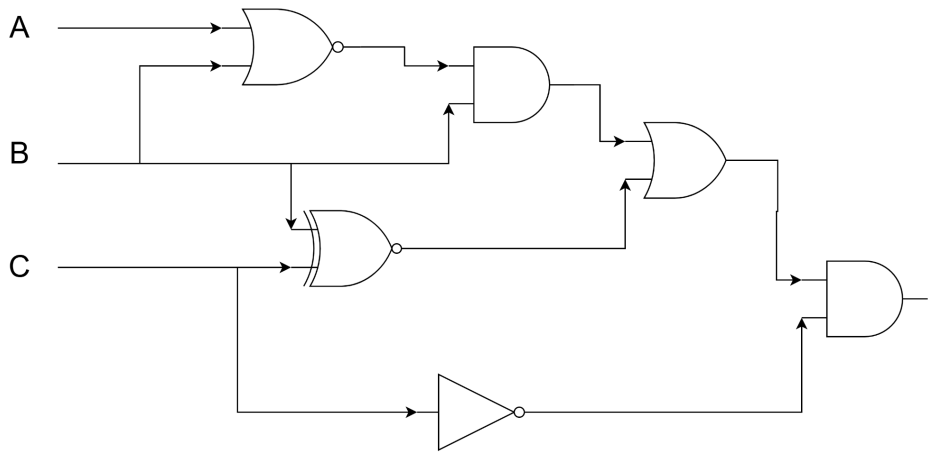
D. 16

2. Boolean Algebra

$A(B + C) + \overline{AB} \oplus \overline{C} = AB + AC + (\overline{A} + \overline{B})C + AB\overline{C}$
$= AB + (A + \overline{A} + \overline{B})C = AB + C$
$A(B + C) + AB \oplus C = AB + AC + AB\overline{C} + (\overline{A} + \overline{B}) * C$
$= AB + (A + \overline{A} + \overline{B})C = AB + C$
Either works because $\overline{AB} \oplus \overline{C} = AB \oplus C$ so you could recognize that and make it easier to keep the negations straight.

C. $AB + C$

3. Digital Electronics



This translates to: $((\overline{A + B})B + \overline{B \oplus C})\,\overline{C}$

D. $((\overline{A + B})B + \overline{B \oplus C})\,\overline{C}$

4. Bit-String Flicking

10011101_2 AND (LSHIFT-3 (RCIRC-2 DB_{16}))
10011101 AND (LSHIFT-3 (RCIRC-2 11011011))
= 10011101 AND (LSHIFT-3 11110110)
= 10011101 AND 10110000
= 10 010 000
= 2 2 0_8

C. 220

5. Bit String Flicking

(RCIRC-2 ((RSHIFT-1 10001) OR X OR (LCIRC-2 10010))) = (LCIRC-3 (NOT 00100))

LHS: (RCIRC-2 (01000 OR abcde OR 01010)) = (RCIRC-2 (01000 OR abcde OR 01010)) = (RCIRC-2 a1c1e) = 1ea1c
RHS: (LCIRC-3 11011) = 11110
Therefore, a=1, b=*, c=0, d=*, e=1 which is 1*0*1.

D. 1*0*1

6. Recursive Functions

Given:

$$f(x) = \begin{cases} f(x-3)+2 & \text{if } x \geq 12 \\ 3 \cdot f(\ \lfloor x \div 2 \rfloor\ ) - 1 & \text{if } 6 \leq x < 12 \\ x-2 & \text{if } x < 6 \end{cases}$$

$f(f(f(f(14))))$ involves the following 4 evaluations:
$f(14) = f(11) + 2 = 8 + 2 = 10$, $f(11) = 3 * f(5) - 1 = 3 * 3 - 1 = 8$, $f(5) = 5 - 2 = 3$
$f(10) = 3 * f(5) - 1 = 3 * 3 - 1 = 8$
$f(8) = 3 * f(4) - 1 = 3 * 2 - 1 = 5$, $f(4) = 4 - 2 = 2$
$f(5) = 3$

E. None of the above

7. Recursive Functions

Given:

$$f(s) = \begin{cases} s & \text{if } len(s) < 2 \\ f(s[len(s)-1:]) + s[0] & \text{otherwise} \end{cases}$$

f(s[len(s)-1:]) performs the function on all but the first character since for a string of length 6, it uses a substring of the last 5 characters.
$f(\text{“}python\text{”}) = f(\text{“}ython\text{”}) + \text{”}p\text{”} = \text{”}nohtyp\text{”}$
$f(\text{“}ython\text{”}) = f(\text{“}thon\text{”}) + \text{”}y\text{”} = \text{”}nohty\text{”}$
$f(\text{“}thon\text{”}) = f(\text{“}hon\text{”}) + \text{”}t\text{”} = \text{”}noht\text{”}$
$f(\text{“}hon\text{”}) = f(\text{“}on\text{”}) + \text{”}h\text{”} = \text{”}noh\text{”}$
$f(\text{“}on\text{”}) = f(\text{“}n\text{”}) + \text{”}o\text{”} = \text{“}no\text{”}$
$f(\text{“}n\text{”}) = \text{“}n\text{”}$
This recursive function reverses a string so the answer is “nohtyp”.

D. nohtyp

8. Prefix/Infix/Postfix Notation

+ / * % 4 + 3 5 % 9 * ^ + ^ 4 2 ^ 3 2 / 1 2 % 5
= + / * 2 8 4 * ^ + 16 9 / 1 2 2
= + 4 * ^ 25 0.5 2
= + 4 * 5 2
= 4 + 5 * 2
= 14

B. 14

9. Prefix/Infix/Postfix Notation

− + ^ 2 3 / * 4 − / 8 2 4 9 / * 6 4 8
= − + (2 3 ^) / * 4 − (8 2 /) 4 9 / (6 4 *) 8
= − + (2 3 ^) / * 4 [(8 2 /) 4 −] 9 [(6 4 *) 8 /]
= − + (2 3 ^) / {4 [(8 2 /) 4 −] *} 9 [(6 4 *) 8 /]
= − + (2 3 ^) ({4 [(8 2 /) 4 −] *} 9 /) [(6 4 *) 8 /]
= − [(2 3 ^) ({4 [(8 2 /) 4 −] *} 9 /) +] [(6 4 *) 8 /]
= 2 3 ^ 4 8 2 / 4 − * 9 / + 6 4 * 8 / −

B. 23^482/4−*9/+64*8/−

10. Computer Number Systems

$1979_{10} = 3673_8$ so counting 48_{10} numbers yields 3673, 3674, 3675, 3676, 3677, 3700-07, 3710-17, 3720-27, 3730-37, 3740-47, 3750, 3751, 3752. There are 9 in 3700-07 and 1each in 3710, 3720, …, 3750 for a total of 14.

C. 14

11. Computer Number Systems

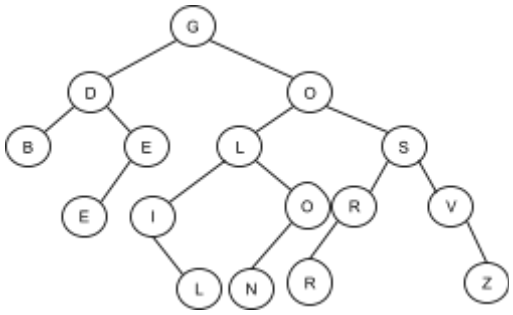
Adding a value of 6 to each component of the color *dark salmon* represented by #E9967A yields E9+6=EF, 96+6=9C, 7A+6=80 which is #EF9C80.

A. #EF9C80

12. Data Structures

The binary search tree to the right represents the string: GOLDSILVERBRONZE.

The preorder traversal of a binary tree uses root, left, right recursively. Therefore, it is GDBEEOLILONSRRVZ.



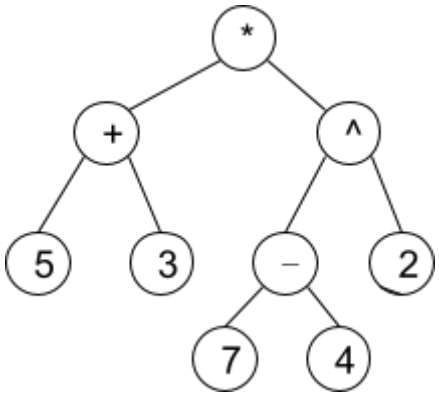
C. GDBEEOLILONSRRVZ

13. Data Structures

Use the following algorithm to build an expression tree:

```
for every character ch:
   if ch is a digit
      t = NEWTREE(nil,ch,nil)
      PUSH(t)
   else
      y = POP()
      x = POP()
      t = NEWTREE(x,ch,y)
      PUSH(t)
```

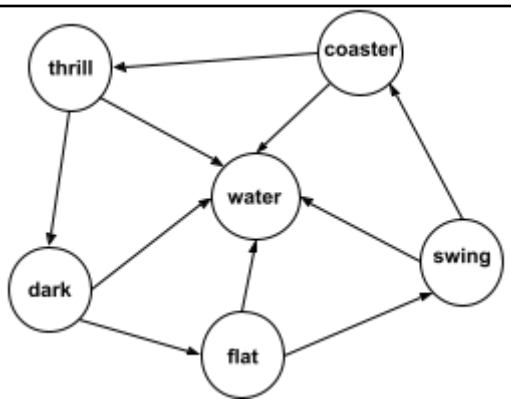
5 3 + 7 4 − 2 ^ * is represented by:



D. d

14. Graph Theory

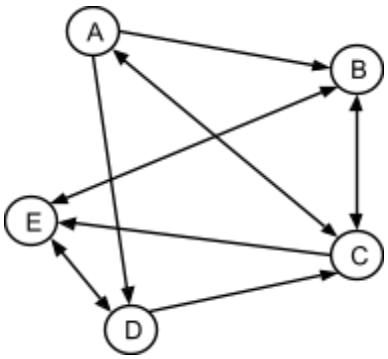
The paths are CTD, CTW, TDW, TDF, DFW, DFS, FSW, FSC, SCT, SCW. There are 10 of them.



C. 10

15. Graph Theory

All of the cycles are: BCB, BEB, BCEB, ACA, ABCA, DED, CEDC, BEDCB, ABEDCA, ADECBA. There are 2 of them that include all 5 vertices.



B. 2

16. What Does This Program Do?

```
for j = 1 to n-1
   key = arr(j)
   k = j - 1
   while k>=0&&key<arr(k)
      [          ]
      k = k - 1
   end while
   arr(k+1) = key
next j
```

This program compares each item but the first with those before it and moves others to make room for it so the first part of the array is sorted. Therefore, the missing statement must be: `arr(k+1)=arr(k)`

C. `arr(k+1)=arr(k)`

17. LISP Programming

To evaluate:
(SETQ X '((a (b b r a) (c a)) ((d a) (b r a))))
(REVERSE (CONS (CDADAR X) (CADADR X)))

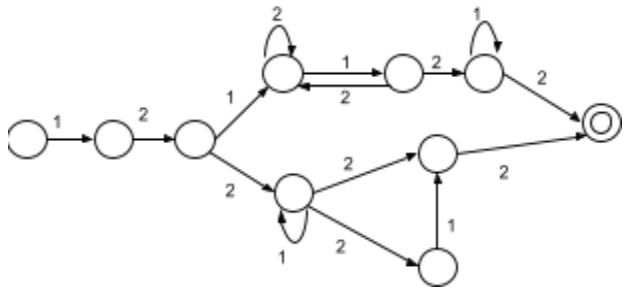
(CDADAR '((a (b b r a) (c a)) ((d a) (b r a))))
= (CDADR '(a (b b r a) (c a)))
= '(b r a)

(CADADR ((a (b b r a) (c a)) ((d a) (b r a))))
= (CADR '((d a) (b r a)))
= '(b r a)

(REVERSE (CONS '(b r a) '(b r a))) = (a r b (b r a))

D. (a r b (b r a))

18. FSAs and Regular Expressions



The strings are:
a.1211222122
b.1222112
c.121122
d.1221111111212
e.121221221212
f.12211121
a, c, d, e are OK, b can't have 2 1s before the 2, and f can't end in a 1

C. a, c, d, e

19. FSAs and Regular Expressions

For the regular expression: [^g-r] [^aeiou] [aei] [i-t]* [a-g]* [s-x]*
a) skiing is OK
b) bobsleigh stops at o with [^aeiou]
c) slalom is OK
d) curling stops at u with [^aeiou]
e) hockey stops at h with [^g-r]
f) snowboard stops at o with [aei]
g) freestyle stops at y with [s-x]*
h) skating is OK

B. a, c, h

20. Assembly Language

What is the output when the following program is executed?

```
NUM     DC      2026
INIT    DC      0
COUNT   DC      1
TOP     LOAD    INIT
        STORE   T
LOOP    LOAD    NUM
        BE      CONT
        DIV     =10
        STORE   QUOT
        MULT    =10
        SUB     NUM
        MULT    =-1
        STORE   D
        MULT    D
        ADD     T
        STORE   T
        LOAD    QUOT
        STORE   NUM
        BU      LOOP
CONT    LOAD    COUNT
        ADD     =1
        STORE   COUNT
        LOAD    T
        STORE   NUM
        SUB     =1
        BE      STOP
        BU      TOP
STOP    PRINT   COUNT
        END
```

The number 2026 is a Happy number because it creates a sequence of the sum of the squares of its digits that ends with 1: 2026, 44, 32, 13, 10, 1. The length of the sequence is 6.

B. 6